

Test: SJChO Past year question

Question 1 of 49

Rutherford bombarded gold foil with alpha particles (An alpha particle consists of two protons and two neutrons bound together into a particle identical to a helium-4 nucleus.) and found that a small percentage of the particles were deflected. Which of the following was not accounted for by the model he proposed for the structure of atoms?

- ☐ A) the total mass of the atom
- ☐ B) the existence of protons
- ☐ C) the charge on the nucleus
- ☐ D) the small size of the nucleus

Question 2 of 49

Silicon, which makes up about 25% of Earth's crust by mass, is used widely in the modern electronics industry. It has three naturally occurring isotopes, ^{28}Si , ^{29}Si , and ^{30}Si . Calculate the atomic mass of silicon. (amu: atomic mass unit)

Isotope	Isotopic Mass (amu)	Abundance %
^{28}Si	27.976927	92.23
^{29}Si	28.976495	4.67
^{30}Si	29.973770	3.10

- ☐ A) 28.9757 amu
- ☐ B) 28.0855 amu
- ☐ C) 29.2252 amu
- ☐ D) 28.7260 amu

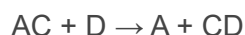
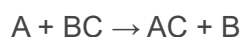
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Carbon-14 is a radioactive isotope which decays with a half-life of 5730 years. What is the first-order rate constant for its decay, in units of years^{-1} ? (For the first order decay process, a half-life ($t_{1/2}$) is calculated as $t_{1/2} = 0.693 / k$, k is the rate constant.)

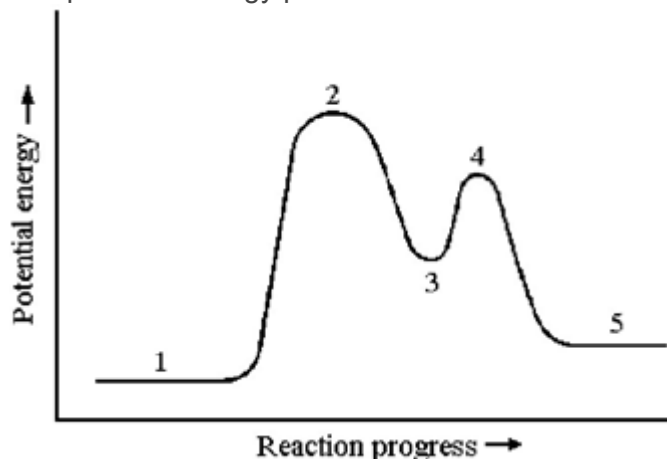
- ☐ A) $1.75 \times 10^{-4} \text{ years}^{-1}$
- ☐ B) $1.21 \times 10^{-4} \text{ years}^{-1}$
- ☐ C) $5.25 \times 10^{-5} \text{ years}^{-1}$
- ☐ D) $3.49 \times 10^{-4} \text{ years}^{-1}$

Question 4 of 49

Consider a reaction that occurs by the following mechanism:



The potential energy profile for this reaction is shown below.



At which reaction stages do intermediates occur?

- ☐ A) reaction stages 2 and 4
- ☐ B) reaction stages 1 and 5
- ☐ C) reaction stage 3
- ☐ D) reaction stages 2, 3, and 4

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Electronegativity is a measure of the tendency of an atom to attract a bonding pair of electrons. The Pauling scale is the most commonly used. Fluorine (the most electronegative element) is assigned a value of 4.0, and values range down to cesium and francium which are the least electronegative at 0.7. Based on the indicated electronegativities, arrange the following in order of increasing ionic character: CsBr, LaBr₃, PBr₃, MgBr₂.

Element	Electronegativity
Br	2.8
P	2.1
Mg	1.2
La	1.0
Cs	0.7

- ☐ A) PBr₃, MgBr₂, LaBr₃, CsBr
- ☐ B) CsBr, MgBr₂, PBr₃, LaBr₃
- ☐ C) PBr₃, LaBr₃, MgBr₂, CsBr
- ☐ D) CsBr, LaBr₃, MgBr₂, PBr₃

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In the Periodic Table atoms are arranged in order of

- ☐ A) Atomic mass
- ☐ B) Periodicity
- ☐ C) Atomic numbers
- ☐ D) Physical properties

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A solution is prepared by adding 0.10 mol of lithium nitrate, LiNO_3 , to 1.00 L of water. Which statement about the solution is correct?

- ☐ A) The solution is neutral.
- ☐ B) The solution is strongly acidic.
- ☐ C) The solution is basic.
- ☐ D) The solution is weakly acidic.

Question 8 of 49

In a report published by two scientists from Harvard University in 2017, metallic hydrogen was successfully produced upon applying a high pressure of 495 gigapascals (GPa) to hydrogen gas. Metallic hydrogen is a liquid made of hydrogen atoms and behaves like a metal.

Which statement about metallic hydrogen is **FALSE**?

- ☐ A) It has the same molar mass as hydrogen gas.
- ☐ B) Turning hydrogen gas to metallic hydrogen is a chemical reaction.
- ☐ C) It is an allotrope of hydrogen.
- ☐ D) It could be highly conductive.

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In a certain metal oxide, the oxidation state of the metal is +1. After reducing X grams of the metal oxide, we obtained Y grams of the metal. What is the molar mass of the metal?

- ☐ A) $8Y/(X-Y)$
- ☐ B) $(X-Y)/8Y$
- ☐ C) $16Y/(X-Y)$
- ☐ D) $(X-Y)/16Y$

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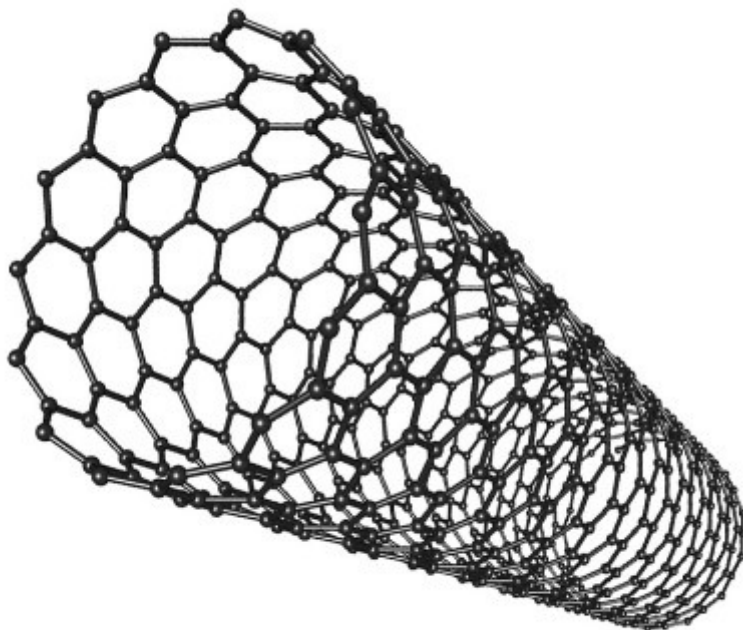
Which of the following liquid has a pH higher than 7?

- (I) Gastric juice
 - (II) Vinegar
 - (III) The aqueous solution of baking soda
 - (IV) Ammonia solution
-

- ☐ **A)** (I), (III), and (IV)
- ☐ **B)** (II) and (III)
- ☐ **C)** (III) only
- ☐ **D)** (III) and (IV)

Question 11 of 49

Carbon nanotubes (CNTs) are cylindrical molecules, whose structure consists of a single layer of carbon atoms, each of which is connected to three neighbouring carbon atoms, as shown in the picture below. CNTs have many attractive features such as ultrahigh strength and low weight.



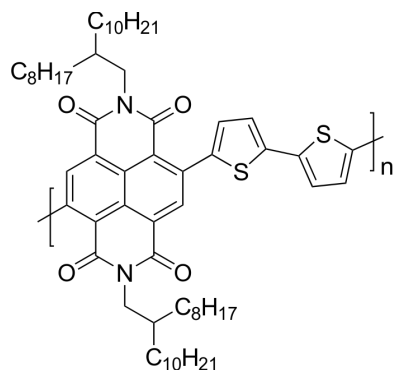
Which of the following statements are true?

-
- ☐ A) The molecular structure and properties of CNTs are similar to those of diamonds.
 - ☐ B) CNTs are polymers.
 - ☐ C) CNTs are stable at room temperature.
 - ☐ D) Conduction of electric current via CNTs is a chemical reaction.

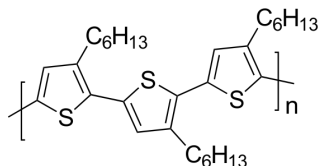
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Organic semiconductors are solid materials that are made of polymers or small molecules, with extended “ π -conjugation” or single-double bond alternations, and are able to conduct electricity when electrons are injected.

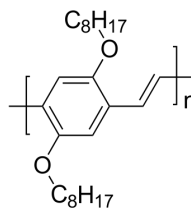
Which of the following polymers are NOT organic semiconductors?



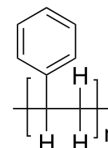
I



II



III

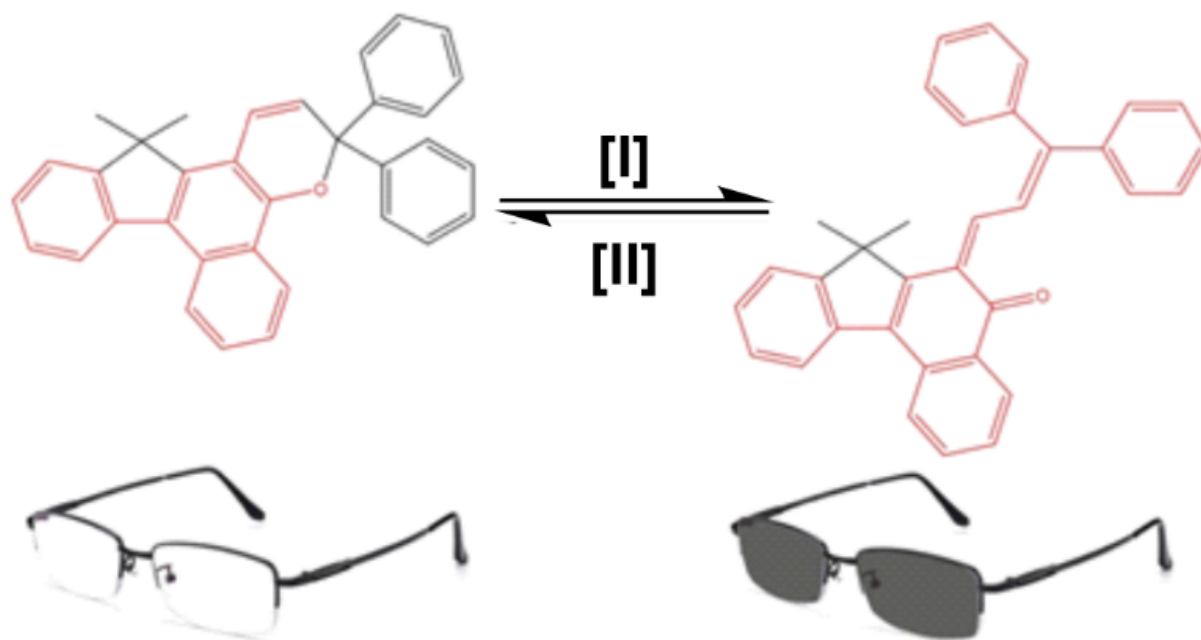


IV

- ☐ A) (IV)
- ☐ B) (II)
- ☐ C) (I), (III), and (IV)
- ☐ D) (III)

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A special type of molecule (photoswitches) is embedded into some sunglasses. This molecule could experience isomerisation as shown below to tune the colour of sunglasses. Under strong sunshine, the colour of the glass turns dark; after entering a room, the glass becomes transparent again. What are the correct reaction conditions [I] and [II] as shown below?



- ☐ A) [I]: heating; [II]: cooling
- ☐ B) [I]: visible light; [II]: dark
- ☐ C) [I]: heating; [II]: dark
- ☐ D) [I]: UV light; [II]: dark

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To prove that a colourless liquid is pure water, which of the following measurement is the most reliable?

- ☐ A) No smell.
- ☐ B) It does not react with sodium (metal).
- ☐ C) Its boiling point is 100°C at 1 atm.
- ☐ D) pH = 7.

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Two burning candles are tightly covered by the glass at the same time. One candle is much taller than the other. Which one will extinguish first?



-
- ☐ A) The short one
 - ☐ B) The tall one
 - ☐ C) It is not possible to predict.
 - ☐ D) Both at the same time

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Penicillin is an important antibiotic that has saved millions of lives. One particular penicillin derivative is a monoacid and forms a salt with either K or Na. For the same amount of this penicillin derivative, the mass of the resulting sodium salt is 12 g, while the mass of the potassium salt is 12.54 g, after complete reactions. What is the relative molecular weight of the penicillin derivative?

- ☐ A) 243.6
- ☐ B) 333.6
- ☐ C) 355.6
- ☐ D) 332.6

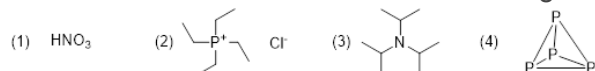
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We live in a colourful world. Which of the following objects' colours are produced by chemical reactions?

- ☐ A) Rainbow
- ☐ B) Colour painting
- ☐ C) Fireworks
- ☐ D) Wings of butterflies

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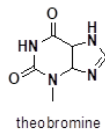
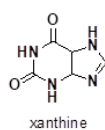
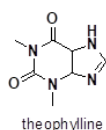
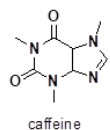
What is the oxidation state of the following Group V atoms in each compound?



- ☐ A) -5, 4, 3, 3
- ☐ B) +5, -3, -3, 0
- ☐ C) +5, 3, 3, 3
- ☐ D) -4, -4, -3, 0

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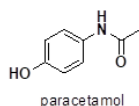
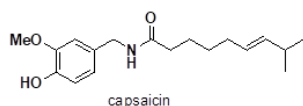
The compounds shown below are natural stimulants that can be found in many foods such as chocolates, tea, and coffee beans. Which of the following statements regarding these compounds is FALSE?



- ☐ A) Theobromine can form dimers by forming hydrogen bonds with another molecule of theobromine.
- ☐ B) Theophylline can form hydrogen-bonds with theobromine.
- ☐ C) Caffeine is expected to be less soluble than xanthine because it can only act as a hydrogen-bond acceptor.
- ☐ D) Xanthine can only act as a hydrogen-bond donor.

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Capsaicin is found in chillis and peppers and is responsible for causing the “hot” and “spicy” sensation when consumed. Paracetamol, also known as acetaminophen, is the active pharmaceutical ingredient found in the drug Panadol that is used to relieve pain and fevers.

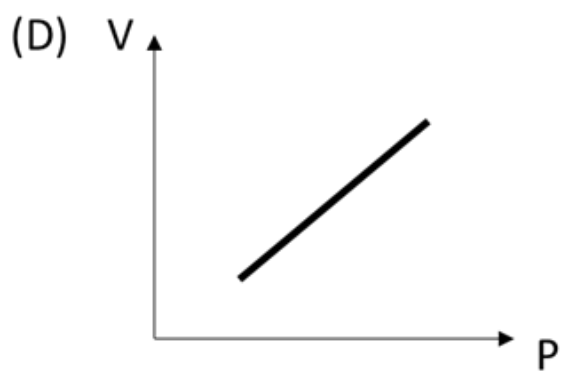
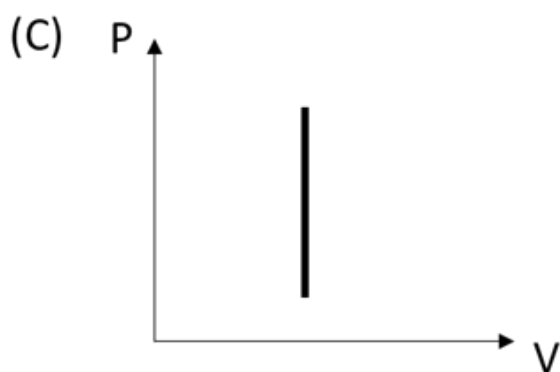
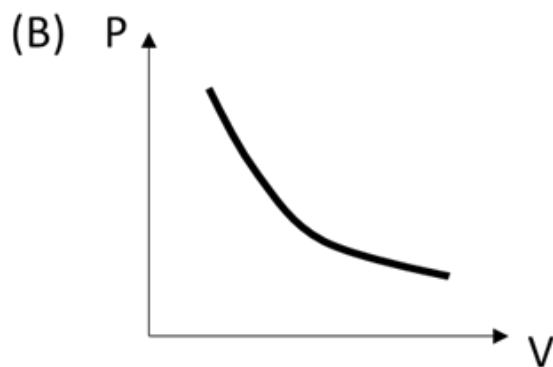
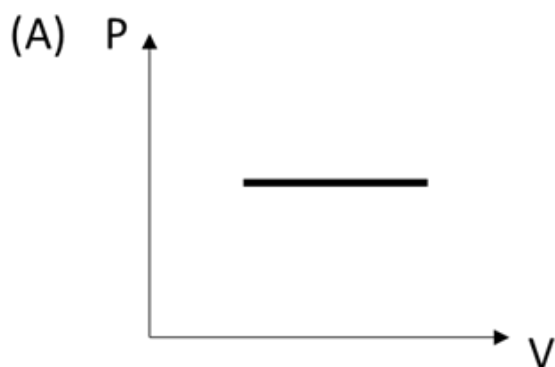


Which one of the following statements about capsaicin and paracetamol is FALSE?

- ☐ A) Capsaicin and paracetamol can undergo hydrolysis.
- ☐ B) Capsaicin can react with hydrogen gas using palladium on carbon as the catalyst.
- ☐ C) When fully dissolved in water, capsaicin forms a solution that has a pH greater than 7.
- ☐ D) Capsaicin and paracetamol can form esters with weak organic acids.

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Which of the following pressure-volume diagram depicts an isochoric process? Isochoric reactions are also known as isovolumetric reactions.

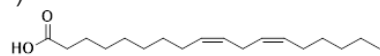


-
- ☐ (A) C
- ☐ (B) A
- ☐ (C) B
- ☐ (D) D

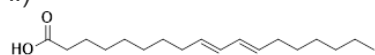
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Fatty acids are carboxylic acids with long hydrocarbon chains. Their melting points are affected by their size, shape and the presence of unsaturated bonds. Which fatty acid below has the lowest melting point?

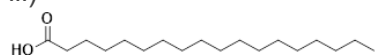
i)



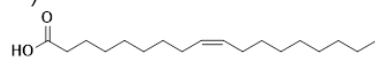
ii)



iii)



iv)



-
- ☐ A) iv
- ☐ B) ii
- ☐ C) i
- ☐ D) iii

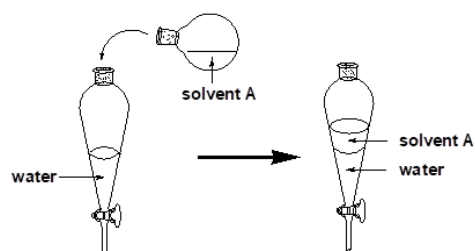
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Iron (III) chloride reacts with hydrogen sulfide to produce iron (III) sulfide and hydrochloric acid. Calculate the mass of iron (III) sulfide produced when 11.61 g of iron (III) chloride reacts with 49.72 g of hydrogen sulfide. Assume the reaction proceeds to completion.

- ☐ A) 7.43 g
- ☐ B) 14.85 g
- ☐ C) 3.71 g
- ☐ D) 38.13 g

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When solvent A in a round bottom flask was added to water in a separating funnel, two layers were formed as shown in the diagram below.



The densities of various solvents are shown below. Which of the following are likely to be solvent A?

	Solvent	Molecular Formula	Density (in g/cm ³)
I	Methylene chloride	CH ₂ Cl ₂	1.33
II	Bromocyclohexane	C ₆ H ₁₁ Br	1.32
III	Cyclohexane	C ₆ H ₁₂	0.81
IV	2-Propanol	C ₃ H ₈ O	0.79
V	Diethyl ether	C ₄ H ₁₀ O	0.71
VI	n-Heptane	C ₇ H ₁₆	0.68

- ☐ A) III, IV, V, VI
- ☐ B) I, II, III, IV
- ☐ C) III, V, VI
- ☐ D) IV, V, VI

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A 28.72 g mineral sample containing $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ was dissolved in distilled water to give a 750 cm^3 solution. 50 cm^3 of the above solution was sent for analysis. The magnesium ions in the 50 cm^3 solution were precipitated as MgNH_4PO_4 . The precipitate was then heated at high temperature to form 0.2965 g of $\text{Mg}_2\text{P}_2\text{O}_7$. What is the percentage by mass of $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ in the mineral sample?

- ☐ A) 8.1
- ☐ B) 14.2
- ☐ C) 1.9
- ☐ D) 28.3

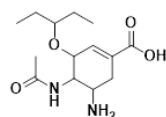
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When compound M was heated in concentrated sulphuric acid, the products obtained were 2-methyl-1-butene and 2-methyl-2-butene. What is compound M? (Hint: this is a dehydration reaction. i.e., one molecule of water is removed)

- ☐ A) 2-methyl-2-butanol
- ☐ B) 3-methyl-1-butanol
- ☐ C) 3-methyl-2-butanol
- ☐ D) 2-methyl-1-butanol

Question 27 of 49

The structure of an antiviral drug is shown below. How many chiral carbon centres does it contain? Chiral carbon centres has 4 different groups linked to them.



- ☐ A) 6
- ☐ B) 4
- ☐ C) 3
- ☐ D) 5

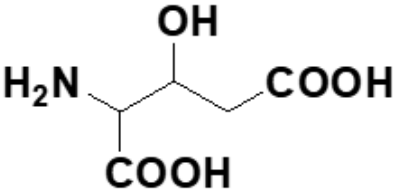
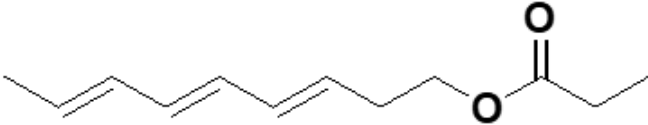
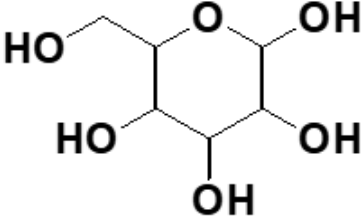
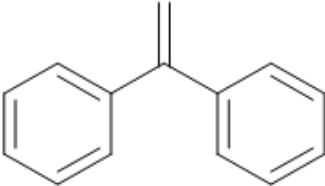
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When compound P is added to sodium carbonate, carbon dioxide is generated. Which of the following could be compound P?

- ☐ A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- ☐ B) $\text{CH}_3\text{CH}_2\text{CHO}$
- ☐ C) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
- ☐ D) $\text{CH}_3\text{CH}_2\text{COOH}$

Question 29 of 49

Reversed phase solid-phase extraction (SPE) is a method to separate mixtures of analytes based on their polarity. The stationary phase of a reversed phase SPE cartridge has long hydrocarbon chains. During reversed phase SPE, which of the analytes below are likely to be retained in the cartridge?

I	
II	
III	
IV	

- ☐ A) I, II, IV
- ☐ B) I, II
- ☐ C) I, III
- ☐ D) II, IV

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When 65.0 cm^3 of solution X was mixed with 375 cm^3 of solution Y, the pH of the final solution was 0.376. $\text{pH} = -\log [\text{H}^+]$. What are solutions X and Y?

- ☐ A) X is $0.25 \text{ mol dm}^{-3} \text{NaCl}$; Y is $0.45 \text{ mol dm}^{-3} \text{HCl}$
- ☐ B) X is $0.25 \text{ mol dm}^{-3} \text{HCl}$; Y is $0.45 \text{ mol dm}^{-3} \text{HCl}$
- ☐ C) X is $0.25 \text{ mol dm}^{-3} \text{H}_2\text{SO}_4$; Y is $0.45 \text{ mol dm}^{-3} \text{HCl}$
- ☐ D) X is $0.25 \text{ mol dm}^{-3} \text{NaOH}$; Y is $0.45 \text{ mol dm}^{-3} \text{HCl}$

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For the following reaction, identify the *oxidant* and the *reductant*.



- ☐ A) Oxidant = $\text{Fe}(\text{OH})_3$ Reductant = I^-
- ☐ B) Oxidant = Fe^{2+} Reductant = I_2
- ☐ C) Oxidant = I^- Reductant = $\text{Fe}(\text{OH})_3$
- ☐ D) Oxidant = H^+ Reductant = $\text{Fe}(\text{OH})_3$

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What is the correct stable structure of the molecule CO_3^{2-} ?

- ☐ A) Triangular with C in the middle and two oxygen atoms attached by a single bond and one by a double bond.
- ☐ B) Linear $[\text{O}-\text{C}-\text{O}-\text{O}]^{2-}$.
- ☐ C) Pyramidal with C on the top with a lone electron pair and oxygen atoms attached to it, two O atoms by single bonds and one by double bond.
- ☐ D) Triangular with C in the middle and all 3 oxygen atoms attached by bonds of the same length.

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Given the reaction:

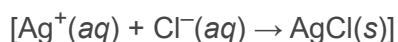


If you started with 5.0 mols of sulfur and you produced 351 g of sulfur trioxide, what is your percent yield? (Molar mass: S= 32.06 g/mol; O₂= 32.00 g/mol; SO₃= 80.06 g/mol.)

-
- ☐ A) 88%
- ☐ B) 93%
- ☐ C) 100%
- ☐ D) 44%

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A chloride of a Group 5A element has the formula **MCl₃**. When 0.100 g of **MCl₃** was added to water and the resulting solution titrated with 0.0500 mol dm⁻³ silver nitrate, it was found that 33.0 cm³ of aqueous silver nitrate was needed to precipitate all the chloride ions. Use these data to calculate the molar mass (*M*) of **MCl₃** and hence identify element **M**.



-
- ☐ A) N
- ☐ B) P
- ☐ C) Sb
- ☐ D) As

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A molten chloride of a Group 1 metal, XCl, and a molten chloride of a Group 2 metal, YCl₂, are separately electrolysed using the same current for the same period of time. Which statement about the experiment is correct?

-
- ☐ A) The masses of elemental X and Y that are deposited are independent of the electrolysis time.
- ☐ B) YCl₂ gives off twice the volume of chlorine gas as XCl, measured under the same conditions of temperature and time.
- ☐ C) An equal number of moles of X and Y are deposited.
- ☐ D) The number of moles of Y deposited is half the number of moles of X deposited.

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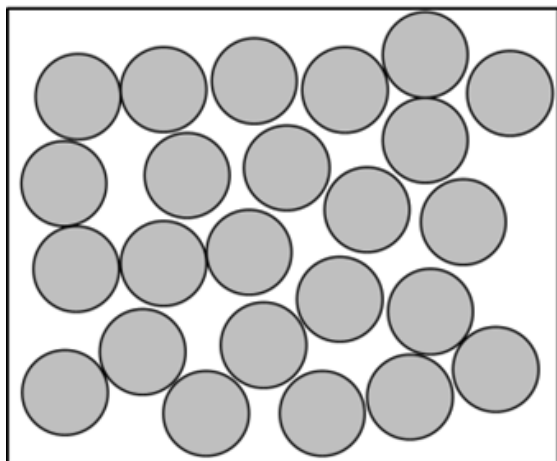
When a copper wire and zinc wire are stuck into a lemon, and the two wires are connected to a light emitting diode (LED), what will happen?

- ☐ A) Light will be generated *via* the reaction of copper wire ($\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$), and *via* a similar reaction of zinc wire ($\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$). The electrons withdrawn from the metals will light the LED.
- ☐ B) No light will be generated. To light the LED, the zinc wire should be connected with the ground (not the lemon) so electrostatic electricity can discharge to ground and in the process it will generate light.
- ☐ C) Light will be generated *via* a spontaneous reaction of the zinc wire ($\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$), which will allow electrons to flow through the LED, generate light, and further flow through the copper wire causing H^+ in the lemon juice to be reduced.
- ☐ D) Light will be generated as the bioelectricity of the lemon will start to flow in the electrical circuit.

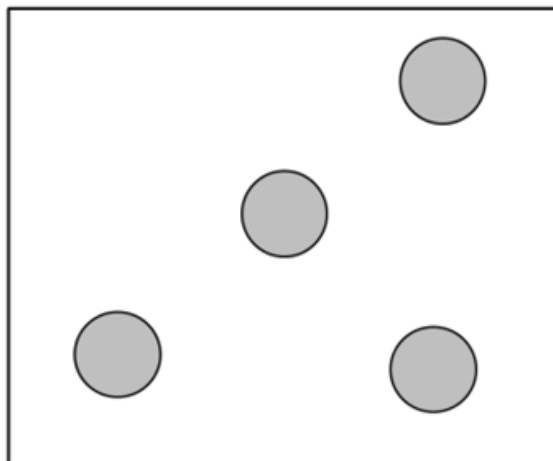
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The diagrams show the arrangement of particles in a substance at two different temperatures. Which of the following substances A to D could be this substance?

-130°C



-30°C



- ☐ A) melting point / °C : -120
boiling point / °C : -25
- ☐ B) melting point / °C : -140
boiling point / °C : -35
- ☐ C) melting point / °C : -140
boiling point / °C : -25
- ☐ D) melting point / °C : -120
boiling point / °C : -35

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Two of the most common nuclear decay reactions are alpha and beta decay. In alpha decay, the atomic nucleus emits an alpha particle or helium nucleus (${}^4_2\text{He}$). Beta decay occurs when one of the neutrons in an atomic nucleus transforms into a proton and an electron. The electron is emitted from the nucleus as a beta particle.

Naturally occurring uranium (${}^{238}_{92}\text{U}$) undergoes a series of 14 radioactive decays to lead-206. The first three decays involve an alpha emission and two beta emissions. What is the nuclide produced after the first three decays?

- ☐ A) ${}^{232}_{88}\text{Ra}$
- ☐ B) ${}^{234}_{91}\text{Pa}$
- ☐ C) ${}^{234}_{92}\text{U}$
- ☐ D) ${}^{234}_{90}\text{Th}$

Question 39 of 49

Aluminium oxide is an ionic compound. In which of the following state/s does it conduct electricity?

- i) Solid
- ii) Molten
- iii) Dissolved in Water

- ☐ A) ii
- ☐ B) i, ii and iii
- ☐ C) iii
- ☐ D) i and ii

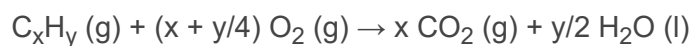
Question 40 of 49

A mixture of $\text{FeSO}_4 \cdot 4\text{H}_2\text{O}$ and $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ is heated to expel the waters of hydration. If 10.00 g of the mixture gives 6.00 g of the anhydrous salts, what is the mass of $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ in the mixture?

- ☐ A) 6.68 g
- ☐ B) 5.38 g
- ☐ C) 4.62 g
- ☐ D) 3.32 g

Question 41 of 49

Gaseous hydrocarbons burn in oxygen according to the following equation:



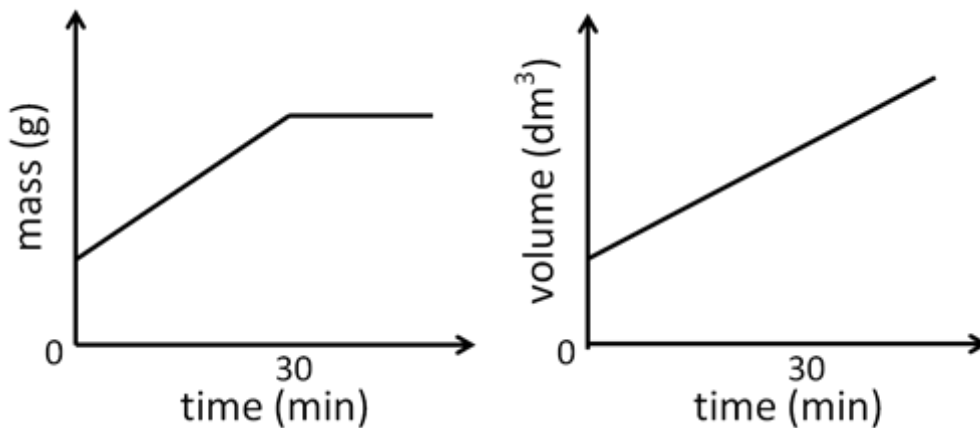
20 cm³ of a gaseous hydrocarbon was fully burnt in 150 cm³ of oxygen gas. The volume of gas remaining at the end of the reaction was 110 cm³. After passing the remaining gas over sodium hydroxide, this volume was reduced to 50 cm³. All volumes were measured at the same temperature and pressure. What is the identity of the hydrocarbon?

-
- ☐ A) C₃H₆
 - ☐ B) C₃H₈
 - ☐ C) C₄H₁₀
 - ☐ D) C₄H₈

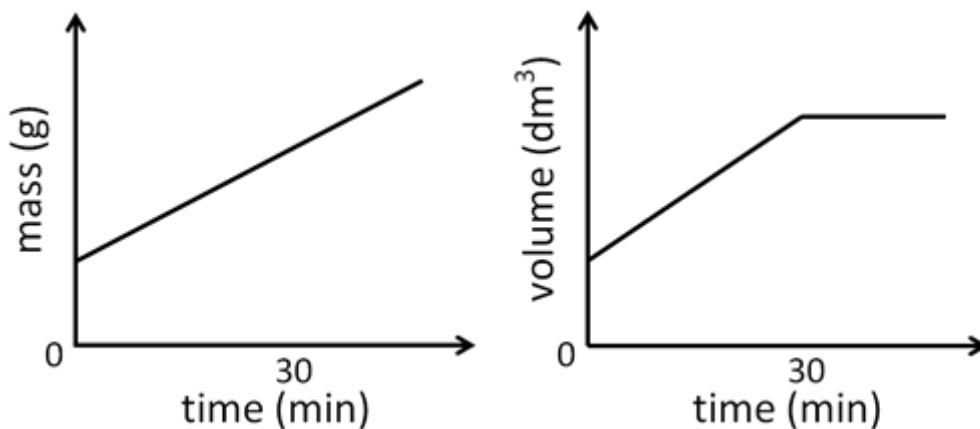
Question 42 of 49

A student performed electrolysis of aqueous copper(II) sulphate using platinum electrodes. He observed that the solution turned colourless after 30 minutes. Which of the following graphs best represent the mass of the cathode and volume of oxygen gas produced at the anode during the electrolysis?

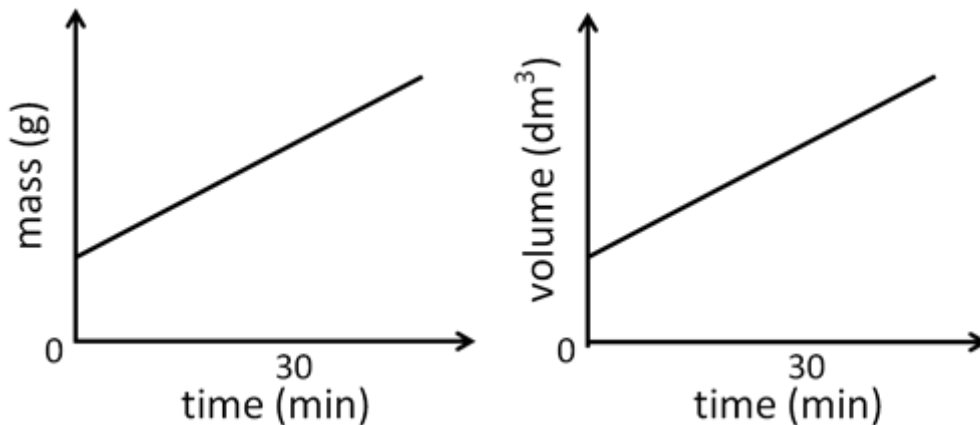
☐ A)



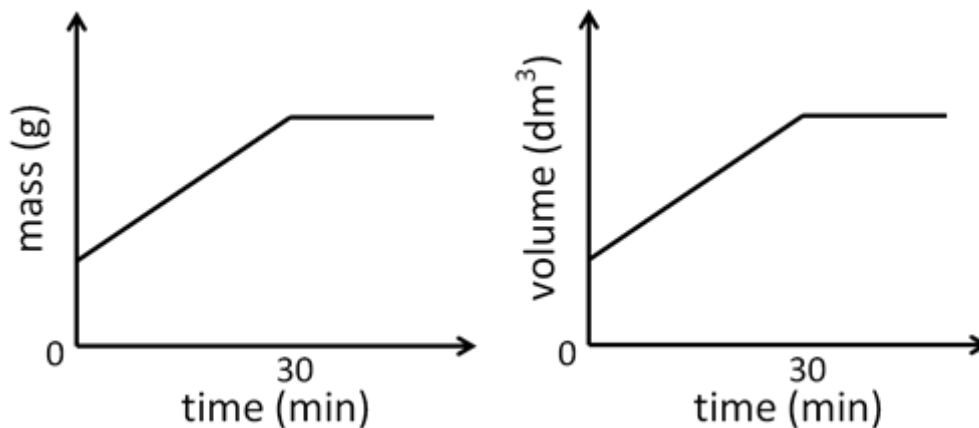
☐ B)



☐ C)



☐ D)



Question 43 of 49

Which of the following processes are endothermic?

- i) Formation of a carbohydrate and oxygen from carbon dioxide and water.
- ii) Production of silver on a photographic film.
- iii) Addition of water to anhydrous copper(II) sulfate.
- iv) Adding concentrated hydrochloric acid to water.

-
- ☐ A) iii and iv
- ☐ B) i, ii, iii and iv
- ☐ C) i and ii
- ☐ D) ii and iii

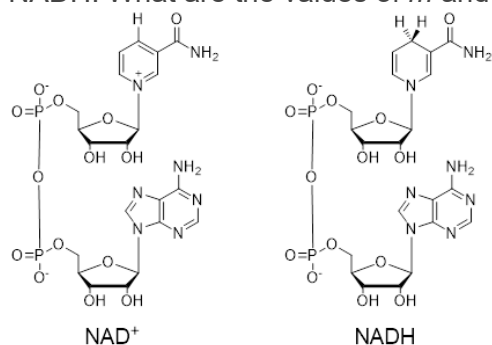
Question 44 of 49

Pure X can be obtained from its oxide by heating with carbon. Its oxide reacts with sodium hydroxide to give a salt and water. Crystals of iron are produced when X is placed in iron(II) sulphate solution. What could be the identity of X?

-
- ☐ A) lead
- ☐ B) copper
- ☐ C) zinc
- ☐ D) aluminium

Question 45 of 49

Nicotinamide adenine dinucleotide is an organic molecule that is involved in redox reactions in enzymes. It exists in two forms: an oxidised form (NAD^+) and a reduced form (NADH). Their structures are shown below. NAD^+ accepts m electrons and $n \text{H}^+$ to become NADH . What are the values of m and n ?



-
- ☐ A) $m: 2$
 $n: 1$
- ☐ B) $m: 2$
 $n: 2$
- ☐ C) $m: 1$
 $n: 1$
- ☐ D) $m: 1$
 $n: 2$

Question 46 of 49

An outline of the Periodic Table is shown below. The positions of eight elements are indicated by the letters A to H.

A	B					D									F	G	
															H	I	
	C																
										E							

Rank the order of reactivity of metals A, B, C, D, and E from most reactive to least reactive.

- ☐ A) E, C, A, B, D
- ☐ B) C, A, B, D, E
- ☐ C) A, C, B, D, E
- ☐ D) A, B, C, D, E

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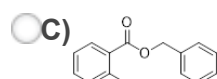
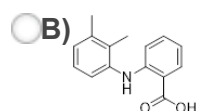
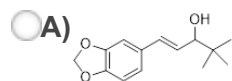
Which statements about carbon monoxide are correct?

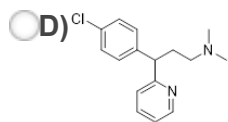
- i) It reacts with sulfur dioxide in catalytic converters to produce harmless products.
- ii) It is one of the waste gases removed by powdered calcium carbonate in coal-burning power stations.
- iii) It is produced when hydrocarbons are burnt in internal combustion engines.
- iv) It is a greenhouse gas.

- ☐ A) iii only
- ☐ B) i, ii, iii, and iv
- ☐ C) i and iii
- ☐ D) i and ii

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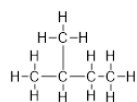
Anne saw a dentist for her toothache and was prescribed painkillers and tablets of aluminium hydroxide. She was told to chew on the tablets before taking the painkillers to prevent stomach discomfort. Which one of the following compounds could be present in the painkillers?





Question 49 of 49

How many possible monosubstituted products are formed when the following alkane reacts with chlorine in the presence of light?



-
- ☐ A) 12
- ☐ B) 5
- ☐ C) 9
- ☐ D) 4